

Business Calculus (MATH 2350)

Summer 2018

Thursday, 06/07/2018

Quiz 4: Section 11.4,11.5

Name (Print): _____

Score: _____ / 100

Section 11.4, 11.5

1.

20 points

 Let y represent a function of x . Derive a general formula for $\frac{d}{dx}(x^n y^m)$. Show all of your steps.

2.

10 points

 Find $\frac{d}{dy}(e^{2y^2+y+1})$.

Hint: Make sure you read the question carefully.

3. The total cost (in hundreds of dollars) of producing x auto body frames is

$$C(x) = 5 + (2x + 3)^{1/3} + 2x.$$

(a) 10 points Find $C'(x)$.

(b) 10 points Find $C'(12)$.

4. For the following problems, find $\frac{dy}{dx}$.

(a) 10 points $x = 4y + 12$

(b) 10 points $\frac{1 + x^2}{1 + y^2} = 1$

(c) 10 points $\ln(x^5) = e^y$

(d) 10 points $x^2 e^y = y + x^2$

(e) 10 points $x^2 + \ln(1/y) = 24$

5. 15 points (bonus) If $f(x) = \frac{x+1}{(x+3)^2}$, find the value(s) of x where the tangent line of f is horizontal.